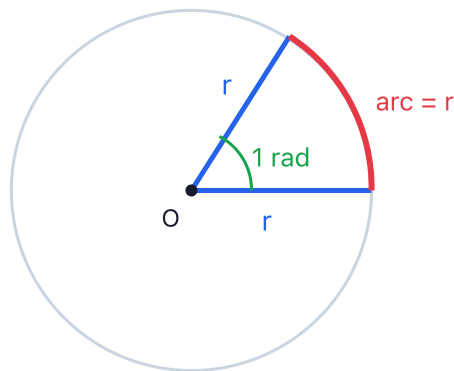


P15: Radians

1. Definition of a Radian

1 radian is the angle subtended at the centre of a circle when the arc length equals the radius.

$$\pi \text{ radians} = 180^\circ$$



When arc length = radius, the angle at centre = 1 radian

2. Converting Between Degrees and Radians

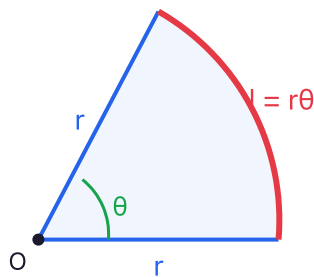
$$\text{Degrees} \rightarrow \text{Radians} : \times \frac{\pi}{180}$$

$$\text{Radians} \rightarrow \text{Degrees} : \times \frac{180}{\pi}$$

Exact Values in Radians

Degrees	Radians	sin	cos	tan
0°	0	0	1	0
30°	$\frac{\pi}{6}$	$\frac{1}{2}$	$\frac{\sqrt{3}}{2}$	$\frac{1}{\sqrt{3}}$
45°	$\frac{\pi}{4}$	$\frac{\sqrt{2}}{2}$	$\frac{\sqrt{2}}{2}$	1
60°	$\frac{\pi}{3}$	$\frac{\sqrt{3}}{2}$	$\frac{1}{2}$	$\sqrt{3}$
90°	$\frac{\pi}{2}$	1	0	undefined
180°	π	0	-1	0

3. Arc Length and Sector Area



$$\text{Arc length: } l = r\theta$$

$$\text{Sector area: } A = \frac{1}{2}r^2\theta$$

$$\text{Segment area: } A = \frac{1}{2}r^2(\theta - \sin \theta)$$

Segment area = Area of sector – Area of triangle

$$A_{\text{segment}} = \frac{1}{2}r^2\theta - \frac{1}{2}r^2 \sin \theta = \frac{1}{2}r^2(\theta - \sin \theta)$$

Perimeter of Sector and Segment

- **Perimeter of sector** = arc length + 2 radii = $r\theta + 2r$
- **Perimeter of segment** = arc length + chord length = $r\theta + 2r \sin\left(\frac{\theta}{2}\right)$

Example: A sector has radius 8 cm and angle $\frac{\pi}{3}$ rad. Find its arc length, area, and perimeter.

$$\text{Arc length: } l = 8 \times \frac{\pi}{3} = \frac{8\pi}{3} \text{ cm}$$

$$\text{Sector area: } A = \frac{1}{2}(8)^2 \times \frac{\pi}{3} = \frac{32\pi}{3} \text{ cm}^2$$

$$\text{Perimeter: } \frac{8\pi}{3} + 2(8) = \frac{8\pi}{3} + 16 \text{ cm}$$

Exam Tip: All arc length and area formulas require θ to be in **radians**. If the angle is given in degrees, convert first.

4. Small Angle Approximations

For small θ (in radians):

$$\sin \theta \approx \theta \quad \cos \theta \approx 1 - \frac{\theta^2}{2} \quad \tan \theta \approx \theta$$

These approximations become more accurate as $\theta \rightarrow 0$. They are derived from the Maclaurin series (not required to derive in exam, but should be able to apply).

Example: Show that for small θ , $\frac{\sin 3\theta}{1 - \cos 2\theta} \approx \frac{3}{2\theta}$.

Numerator: $\sin 3\theta \approx 3\theta$

Denominator: $1 - \cos 2\theta \approx 1 - \left(1 - \frac{(2\theta)^2}{2}\right) = 2\theta^2$

Ratio: $\frac{3\theta}{2\theta^2} = \frac{3}{2\theta}$

Note: This gives $\frac{3}{2\theta}$ as required — always work through the algebra carefully.

Exam Tip: When using small angle approximations, substitute immediately without rearranging first. Always check which approximation applies to which trig function in the expression.